

JEE MAIN REVISION MASTER: THERMODYNAMICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 224 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Enthalpy, Hess's Law & Bond Energies	134 PYQs	Thermochemistry
Very High	Entropy & Gibbs Free Energy Spontaneity	94 PYQs	Spontaneity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Enthalpy, Hess's Law & Bond Energies	$\Delta H = \Delta U + \Delta n_g RT, \quad \Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$
Entropy & Gibbs Free Energy Spontaneity	$\Delta G = \Delta H - T\Delta S, \quad \Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0, \quad \Delta G^\circ = -RT \ln K$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Enthalpy, Hess's Law & Bond Energies (Very High Yield — 134 PYQs)

Core Principle: State functions, standard enthalpy of formation, Hess's law of constant heat summation, and bond energy calculations.

■ **Caution / JEE Pitfall:** Standard enthalpy of formation $\Delta_f H^\circ$ of pure elements in their standard reference state is strictly ZERO (e.g. $O_2(g)$, $C(\text{graphite})$).

■ **Speed Trick:** When using bond enthalpies: $\Delta H_{\text{reaction}} = \sum \text{BE}(\text{bonds broken in reactants}) - \sum \text{BE}(\text{bonds formed in products})$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "20.0 dm³ of an ideal gas 'X' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa. Whi..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Thermodynamic Process List-II Magnitude in kJ A. Work done in reversible, isothermal expansion of 2 m..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "If 3.365 g of ethanol (l) is burnt completely in a bomb calorimeter at 298.15 K, the heat produced is 99.472 kJ. The $\Delta_f H^\circ$ of..." [Ans: 3]

3.2 Entropy & Gibbs Free Energy Spontaneity (Very High Yield — 94 PYQs)

Core Principle: Second and third laws of thermodynamics, entropy changes in phase transitions, and Gibbs criteria for spontaneous change.

■ **Caution / JEE Pitfall:** For an endothermic reaction ($\Delta H > 0$) with $\Delta S > 0$, the reaction is spontaneous ONLY at HIGH temperatures where $T > (\Delta H / \Delta S)$.

■ **Speed Trick:** At equilibrium: $\Delta G = 0$ implies $T_{\text{(equilibrium)}} = (\Delta H / \Delta S)$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Match List - I with List - II. Given V_1 and V_2 are initial and final volumes respectively. List - I (Isothermal process) List - II..." **[Ans: C]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#2 (Numerical):** "At the transition temperature T , $A \ln B$ and $\Delta G^\circ = 105 - 35 \log T$ where A and B are two states of substance X . The ..." **[Ans: 727]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Match List - I with List - II. Given V_1 and V_2 are initial and final volumes respectively. List - I (Isothermal process) List - II (Expression) A. Reversible expansion I. $q=0$ B. Free expansion II. $q=nRT \ln \frac{V_2}{V_1}$ C. Irreversible Compression III. $w=-p_{\text{ext}}(V_1-V_2)$ D. Cyclic reversible IV. $\int_{\text{rev}} \frac{dq}{T}=0$ Choose the correct answer from the options given below :

- (A) -II, B-III, C-I, D-IV
- (B) -II, B-I, C-IV, D-III
- (C) -II, B-I, C-III, D-IV **[Correct]**
- (D) -I, B-II, C-III, D-IV

Correct Answer: C

Solution Analysis: (A) Reversible isothermal expansion For an ideal gas in a reversible and isothermal expansion: $w=-nRT \ln \frac{V_2}{V_1}$ Since the process is isothermal, $\Delta U=0$ ■ $q=-w$ Hence, $q=nRT \ln \frac{V_2}{V_1}$ Therefore, A corresponds to II. (B) Free expansion In free expansion, gas expands against zero external pressure: $p_{\text{ext}}=0$ Thus, $w=0$ and $q=0$ (Also, for an ideal gas, $\Delta U=0$.) Therefore, B corresponds to I. (C) Irreversible isothermal compression For irreversible compression (against constant external pressure): $w=-p_{\text{ext}}(V_1-V_2)$ Therefore, C corresponds to III. (D) Cyclic reversible For a cyclic process, $\int \frac{dq}{T}=0$ Therefore, D corresponds to IV.

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

20.0 dm³ of an ideal gas 'X' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa. Which of the following option is correct? (Given: $\log 2 = 0.3010$ and $\log 5 = 0.6989$)

- (A) $w = -9.1 \text{ kJ}$, $\Delta U = 0$, $\Delta H = 0$, $q = 9.1 \text{ kJ}$ [Correct]
- (B) $w = 9.1 \text{ J}$, $\Delta U = 9.1 \text{ J}$, $\Delta H = 0$; $q = 0$
- (C) $w = -3.9 \text{ kJ}$, $\Delta U = 0$, $\Delta H = 0$; $q = 3.9 \text{ kJ}$
- (D) $w = +4.1 \text{ kJ}$, $\Delta U = 0$, $\Delta H = 0$; $q = -4.1 \text{ kJ}$

Correct Answer: A

Solution Analysis: For an ideal gas undergoing isothermal reversible expansion : Temperature is constant $\Rightarrow \Delta U = 0$ (for ideal gas, U depends only on T) Hence $\Delta H = 0$ (since H also depends only on T for ideal gas) Step 1: Given data (convert units) $V_1 = 20.0 \text{ dm}^3 = 20.0 \times 10^{-3} \text{ m}^3 = 0.020 \text{ m}^3$ $P_1 = 0.5 \text{ MPa} = 0.5 \times 10^6 \text{ Pa} = 5.0 \times 10^5 \text{ Pa}$ $P_2 = 0.2 \text{ MPa}$ $T = 600 \text{ K}$ Step 2: Find number of moles using $PV = nRT$ $n = \frac{P_1 V_1}{RT} = \frac{(5.0 \times 10^5)(0.020)}{(8.314)(600)} = \frac{10000}{4988.4} \approx 2.0$